

Review:

# Smooth and Time-Optimal Trajectory Planning for Industrial Manipulators Along Specified Paths\*

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## Abstract

This report presents a review and implementation-oriented study of Smooth Path-Constrained Time-Optimal Motion (SPCTOM) for industrial manipulators. Classical Path-Constrained Time-Optimal Motion (PCTOM) methods produce bang-bang or bang-singular-bang torque profiles that are time-optimal but discontinuous, leading to large jerk, tracking errors, actuator stress, and vibrations. SPCTOM incorporates actuator torque-rate constraints to generate smooth trajectories while retaining near-optimal traversal time. These constraints better characterize practical actuator limitations and improve robustness against structural flexibility and modelling inaccuracies. The problem is reformulated in the phase plane through path parametrization and solved as a nonlinear optimization problem using spline-based trajectory parameterization and numerical optimization techniques. The framework is also implemented for the standard 1-DoF car problem to analyze representative results. Key implementation challenges, limitations, and future directions are discussed.

## 1 Introduction

Industrial manipulators are required to operate with high productivity while maintaining reliability and minimizing operational cost. Consequently, the computation of time-optimal trajectories is an important problem in robotics and automation.

In many industrial applications, the geometric path of the manipulator is predefined. The problem therefore reduces to finding the fastest feasible traversal along the specified path while respecting actuator and dynamic constraints. This problem is known as Path-Constrained Time-Optimal Motion (PCTOM).

Classical PCTOM solutions typically exhibit bang-bang or bang-singular-bang control structures. Although these trajectories are time-optimal, they contain discontinuities in actuator torque and impulsive jerk. In practice, actuators cannot instantaneously track such discontinuous commands, leading to tracking errors, vibrations, and increased mechanical stress. Persistent excitation of the feedback controller may further accelerate actuator wear and reduce system lifetime.

Several approaches have previously been proposed to smooth discontinuous controls, including modified cost functions involving actuator torques and spline-based trajectory parameterization. While these methods improve smoothness, they often sacrifice traversal time, rely on conservative bounds, or do not explicitly incorporate actuator dynamics and singular controls.

To address these issues, Smooth Path-Constrained Time-Optimal Motion (SPCTOM) incorporates actuator torque-rate constraints into the formulation. By constraining jerk indirectly through torque-rate limits, smoother and more practically realizable trajectories can be obtained. These constraints also better capture practical actuator limitations and help compensate for structural flexibility and modelling inaccuracies, making SPCTOM trajectories more suitable for implementation on industrial robotic systems.

## 2 Smooth Path-Constrained Time Optimal Motion

### 2.1 Problem Formulation

The objective is to determine a trajectory that minimizes traversal time while satisfying manipulator dynamics, actuator constraints, and the prescribed geometric path. The optimization problem is formulated as

$$\min_{\dot{T} \in \Omega} \int_0^{t_f} 1 \, dt$$

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subject to the manipulator dynamics

$$M(q)\ddot{q} + \dot{q}^T C(q, \dot{q})\dot{q} + G(q) = T$$

where:

|                 |                                  |
|-----------------|----------------------------------|
| $M(q)$          | Inertia matrix                   |
| $C(q, \dot{q})$ | Coriolis and centrifugal effects |
| $G(q)$          | Gravity vector                   |
| $q$             | Joint positions                  |
| $T$             | Actuator torques                 |

The boundary conditions are

$$q(0) = q_0, \quad q(t_f) = q_f$$

$$\dot{q}(0) = \dot{q}(t_f) = 0$$

$$\ddot{q}(0) = \ddot{q}(t_f) = 0$$

The actuator torque and torque-rate limits are

$$T_{\min} \leq T \leq T_{\max}$$

$$\dot{T}_{\min} \leq \dot{T} \leq \dot{T}_{\max}$$

The manipulator motion is additionally constrained to follow a prescribed geometric path

$$r = r(s)$$

where  $r(s)$  is a  $C^1$ -continuous curve parametrized by the scalar path variable  $s$ , which may represent the arc length along the path.

In this formulation, the actuator torque rates  $\dot{T}$  constitute the bounded control inputs. Since the standard Lagrangian dynamics explicitly contain only actuator torques  $T$ , the system dynamics must be differentiated with respect to time to obtain a formulation in terms of torque-rate controls. Differentiating the manipulator dynamics yields the third-order system

$$M(q)\ddot{\ddot{q}} + \dot{M}(q)\ddot{q} + \ddot{q}^T C(q, \dot{q})\dot{q} + \dot{q}^T \dot{C}(q, \dot{q})\dot{q} + \dot{q}^T C(q, \dot{q})\ddot{q} + \dot{G}(q) = \dot{T}$$

Equation above is taken as the system dynamics, with  $\dot{T}$  representing the bounded  $n$ -dimensional control input.

Rewriting in state-variable representation, we get:

$$\dot{x} = F(x, u) = F(x) + K(x)u$$

$$x^T = (q^T, \dot{q}^T, \ddot{q}^T), \quad u = \dot{T}$$

where:

$$F(x) = \begin{bmatrix} \dot{q} \\ \ddot{q} \\ -M^{-1}(q) \left( \dot{M}(q)\ddot{q} + \ddot{q}^T C(q, \dot{q})\dot{q} + \dot{q}^T \dot{C}(q, \dot{q})\dot{q} + \dot{q}^T C(q, \dot{q})\ddot{q} + \dot{G}(q) \right) \end{bmatrix}$$

$$K(x) = \begin{bmatrix} 0 \\ 0 \\ M^{-1}(q) \end{bmatrix}$$

$$x_0 = (q_0, 0, 0), \quad x_f = (q_f, 0, 0)$$

## 2.2 Path Constraints

The scalar path parameterization reduces the dimensionality of the trajectory planning problem from  $3n$  to 3, where  $n$  is the number of degrees of freedom of the manipulator. The joint coordinates are expressed as a function of a scalar path parameter  $s$ :

$$q = q(s)$$

Using the chain rule, the manipulator dynamics can be rewritten in terms of the pseudo-velocity  $\dot{s}$ , pseudo-acceleration  $\ddot{s}$ , and pseudo-jerk  $\ddot{\ddot{s}}$ .

The actuator torque constraints become

$$T = A(s)\ddot{s} + B(s)\dot{s}^2 + C(s)$$

subject to

$$T_{\min} \leq T \leq T_{\max}$$

where

$$\begin{aligned} A(s) &= Mq' \\ B(s) &= Mq'' + q'^T Cq' \\ C(s) &= G \end{aligned}$$

Similarly, differentiating the torque equation with respect to time yields the torque-rate constraints

$$\dot{T} = a(s)\ddot{\ddot{s}} + b(s)\dot{s}\ddot{s} + c(s)\dot{s}^3 + d(s)\dot{s}$$

subject to

$$\dot{T}_{\min} \leq \dot{T} \leq \dot{T}_{\max}$$

where

$$\begin{aligned} a(s) &= Mq' \\ b(s) &= 3Mq'' + \frac{dM}{ds}q' + 2q'^T Cq' \\ c(s) &= Mq''' + \frac{dM}{ds}q'' + q''^T Cq' + q'^T \frac{dC}{ds}q' + q'^T Cq'' \\ d(s) &= \frac{dG}{ds} \end{aligned}$$

The resulting reduced-order system has state variables

$$X = (s, \dot{s}, \ddot{s})^T$$

and control input

$$u = \ddot{s}$$

with dynamics

$$\dot{X} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} X + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

Thus, the original manipulator dynamics are transformed into a scalar trajectory planning problem in the  $s$ - $\dot{s}$  phase plane.

### 2.3 Torque and Torque-Rate Constraints

For each manipulator joint, the torque limits consist of an upper and lower bound:

$$T_{i,\min} \leq T_i \leq T_{i,\max}$$

Substituting the path-parametrized dynamics into the above expression yields two linear inequalities in the  $(\ddot{s}, \dot{s}^2)$  plane for every joint. Consequently, an  $n$ -joint manipulator generates  $2n$  boundary lines in this plane. The intersection of all admissible half-planes forms a convex polygonal feasible region, as illustrated in Fig. 1.

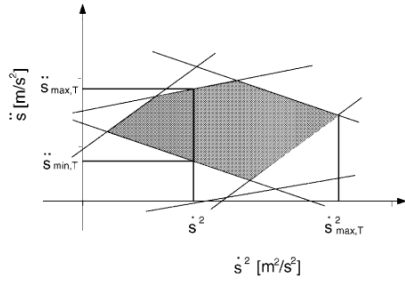


Figure 1: Admissible polygonal region induced by actuator torque limits in the  $(\ddot{s}, \dot{s}^2)$  plane.

For a fixed path position  $s$ , the feasible polygon determines allowable pseudo-acceleration values for a given pseudo-velocity. The admissible pseudo-acceleration bounds are obtained as

$$\ddot{s}_{\min}^T(s, \dot{s}) = \max_i (\ddot{s}_{i,\min}^T(s, \dot{s}))$$

$$\ddot{s}_{\max}^T(s, \dot{s}) = \min_i (\ddot{s}_{i,\max}^T(s, \dot{s}))$$

The maximum feasible pseudo-velocity occurs when the admissible region collapses to a single point, producing the velocity limit curve (VLC):

$$\dot{s} \leq \dot{s}_{\max}^T(s)$$

The VLC therefore represents the maximum achievable path velocity under actuator torque limits.

Similarly, the torque-rate limits for each joint are given by

$$\dot{T}_{i,\min} \leq \dot{T}_i \leq \dot{T}_{i,\max}$$

Substituting the torque-rate dynamics produces two linear inequalities per joint in the  $(\ddot{s}, \dot{s})$  plane. The intersection of these constraints again forms a polygonal admissible region, shown in Fig. 2.

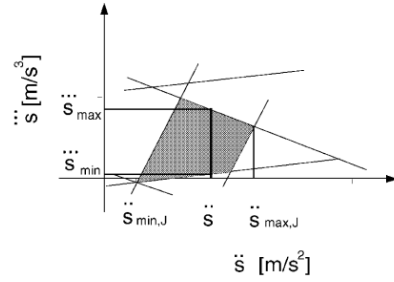


Figure 2: Admissible polygonal region induced by torque-rate limits in the  $(\ddot{s}, \dot{s})$  plane.

From these regions, the admissible pseudo-acceleration and pseudo-jerk limits are obtained as

$$\ddot{s}_{\min}^J(s, \dot{s}) = \max_i (\ddot{s}_{i,\min}^J(s, \dot{s}))$$

$$\ddot{s}_{\max}^J(s, \dot{s}) = \min_i (\ddot{s}_{i,\max}^J(s, \dot{s}))$$

and

$$\ddot{\ddot{s}}_{\min}^J(s, \dot{s}, \ddot{s}) = \max_i (\ddot{\ddot{s}}_{i,\min}^J(s, \dot{s}, \ddot{s}))$$

$$\ddot{\ddot{s}}_{\max}^J(s, \dot{s}, \ddot{s}) = \min_i (\ddot{\ddot{s}}_{i,\max}^J(s, \dot{s}, \ddot{s}))$$

The torque-rate constraints also induce an additional velocity bound

$$\dot{s} \leq \dot{s}_{\max}^J(s)$$

which occurs at points where

$$\ddot{s}_{\min}^J = \ddot{s}_{\max}^J$$

Combining the torque and torque-rate constraints yields the overall feasible pseudo-acceleration bounds

$$\ddot{s}_{\min}(s, \dot{s}) = \max(\ddot{s}_{\min}^T, \ddot{s}_{\min}^J)$$

$$\ddot{s}_{\max}(s, \dot{s}) = \min(\ddot{s}_{\max}^T, \ddot{s}_{\max}^J)$$

The corresponding global velocity bound is

$$\dot{s}_{\max}(s) = \min(\dot{s}_{\max}^T(s), \dot{s}_{\max}^J(s))$$

This final limiting curve in the  $s$ - $\dot{s}$  phase plane is referred to as the Smooth Motion Velocity Limit Curve (SMVLC). The feasible SPCTOM trajectory must remain entirely within this admissible region while satisfying all pseudo-velocity, pseudo-acceleration, and pseudo-jerk constraints.

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### 3 Methodology

Time-optimal control problems similar to SPCTOM are traditionally addressed using Pontryagin’s Maximum Principle (PMP), two-point boundary value problem (TPBVP) formulations, dynamic programming, or specialized switching-point algorithms. However, dynamic programming becomes computationally intractable due to exponential scaling with system dimension. Furthermore, the inclusion of both state and control constraints allows them to become active independently, preventing the optimal control from exhibiting a simple bang-bang or bang-singular-bang structure. Consequently, classical PMP-based approaches are not directly applicable to the SPCTOM problem.

To address these difficulties, the problem is reformulated in the  $s$ - $\dot{s}$  phase plane as a nonlinear parameter optimization problem. This reparameterization fixes the trajectory endpoints in the path domain, unlike the original time-domain formulation where the final time remains free. The traversal time is therefore expressed as

$$T_f = \int_{s_0}^{s_f} \frac{1}{\dot{s}(s)} ds$$

with the pseudo-velocity  $\dot{s}(s)$  and endpoint slopes  $\frac{d\dot{s}}{ds}$  as the optimization variables.

#### 3.1 Trajectory Parameterization

The trajectory in the  $s$ - $\dot{s}$  phase plane is inherently infinite-dimensional. To obtain a finite-dimensional optimization problem, the pseudo-velocity profile is approximated using cubic splines defined over a fixed set of knot points.

Cubic splines are selected because they provide  $C^2$ -continuous trajectories, differentiability required by the dynamics and sufficient flexibility to accurately approximate smooth trajectories.

Higher-order polynomials are avoided since they are unnecessary for the required smoothness and may introduce undesirable oscillatory behavior.

The spline parameterization is defined using the pseudo-velocity values  $\dot{s}$  at the knot points and the boundary slopes  $\frac{d\dot{s}}{ds}$  at the initial and final points.

The knot locations are fixed using switching points obtained from the classical PCTOM solution. This provides an effective initialization since SPCTOM approaches the PCTOM solution as the torque-rate limits become large.

#### 3.2 Optimization Strategy

The resulting optimization problem is nonlinear and non-convex, with no closed-form expressions for the derivatives of either the objective function or the feasibility constraints. Consequently, a numerical evaluation-based optimization strategy is adopted.

The optimization variables are normalized using the PCTOM solution:

$$\mathbf{x} = \left( \frac{\frac{d\dot{s}}{ds}|_0}{\frac{d\dot{s}}{ds}|_{m_0}}, \frac{\dot{s}_1}{\dot{s}_{m_1}}, \frac{\dot{s}_2}{\dot{s}_{m_2}}, \dots, \frac{\dot{s}_p}{\dot{s}_{m_p}}, \frac{\frac{d\dot{s}}{ds}|_f}{\frac{d\dot{s}}{ds}|_{m_f}} \right)^T$$

where the denominator terms correspond to quantities obtained from the PCTOM solution.

Since the optimal trajectory typically lies on the boundary of the feasible region, the optimization method must effectively explore both feasible and infeasible regions while converging toward the feasibility boundary.

To achieve this, the framework employs a Flexible Tolerance Method (FTM) and Nelder-Mead optimization. In FTM, constraints are initially enforced with relaxed tolerances that are progressively tightened during optimization. This allows the search process to move through mildly infeasible regions during early iterations and gradually converge toward a feasible near-optimal solution.

The Nelder-Mead method is used as the underlying numerical optimizer due to its derivative-free formulation, which is suitable for problems where analytical gradients are unavailable and the constraints must be evaluated numerically.

## 4 Results

The SPCTOM framework produces trajectories with significantly smoother pseudo-velocity and torque-rate profiles compared to classical PCTOM solutions. Typical optimization results are shown in Fig. 3 and Fig. 4.

As shown in Fig. 3, the SPCTOM trajectory remains within the admissible region defined by the velocity limit curve while producing a significantly

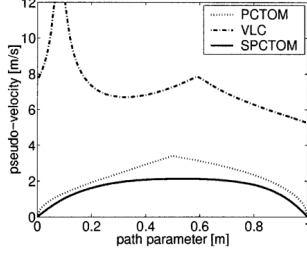


Figure 3: SPCTOM yields smoother pseudo-velocity profiles compared to PCTOM.

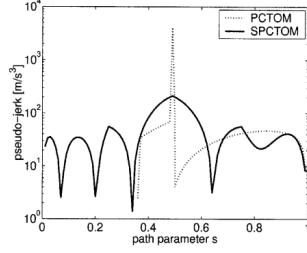


Figure 4: Torque-rate profile in SPCTOM is smooth and continuous.

smoother pseudo-velocity profile than classical PCTOM. In contrast to the sharp switching behaviour typically observed in PCTOM solutions, the SPCTOM trajectory transitions smoothly between acceleration and deceleration phases.

Figure 4 further highlights the effect of torque-rate constraints. Classical PCTOM trajectories exhibit discontinuities and impulsive jerk due to abrupt torque switching. By incorporating torque-rate limits, the SPCTOM framework generates continuous and smooth torque-rate profiles without discontinuities. This reduces excitation of structural vibrations and improves practical realizability on physical manipulators.

#### 4.1 Standard 1-DoF Car Problem

To validate the implementation, the framework was applied to the standard 1-DoF car problem with bounded acceleration and deceleration. This problem is commonly used as a benchmark since the time-optimal solution is analytically known to exhibit bang-bang behaviour.

The classical PCTOM solution accelerates at maximum allowable acceleration and switches instantaneously to maximum braking, producing discontinuous control inputs and impulsive jerk. In contrast, the SPCTOM solution produces smoother transitions between acceleration and deceleration while satisfying the imposed jerk constraints.

As observed in Fig. 5, the SPCTOM trajectory exhibits smooth but slightly oscillatory behaviour due to the spline parameterization and numerical

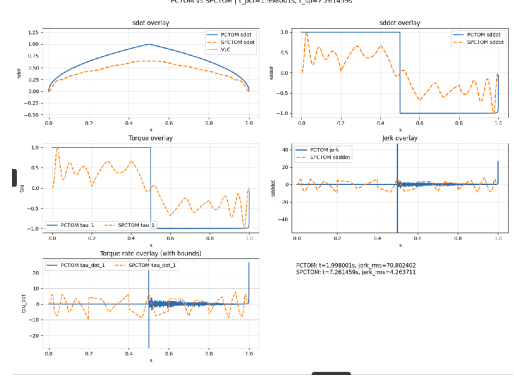


Figure 5: Comparison of PCTOM and SPCTOM trajectories for the standard 1-DoF car problem.

optimization process. While the classical PCTOM solution completes the motion in approximately 2s, the SPCTOM trajectory requires approximately 7s due to the additional smoothness constraints and conservative optimization behaviour.

Although the SPCTOM trajectory is slower in this benchmark example, it eliminates impulsive torque behaviour and generates dynamically smoother motion. Further discussion regarding tracking performance, practical feasibility, and optimization trade-offs is provided in the evaluation section.

## 5 Evaluation

The proposed SPCTOM framework is evaluated in terms of trajectory smoothness, tracking feasibility, execution time, and robustness of the optimization process. Comparative results between classical PCTOM and SPCTOM trajectories are summarized in Fig. 6.

| Trajectory | Torque rate limits [Nm/s] | Motion time [s] | Maximum tracking error [cm] | RMS tracking error |             |             |
|------------|---------------------------|-----------------|-----------------------------|--------------------|-------------|-------------|
|            |                           |                 |                             | joint 1 [°]        | joint 2 [°] | joint 3 [°] |
| PCTOM      | $\infty$                  | 0.90            | 1.98                        | 1.54               | 0.31        | 0.37        |
| SPCTOM 2   | 100                       | 0.74            | 1.40                        | 1.12               | 0.26        | 0.33        |
| SPCTOM 3   | 10                        | 1.50            | 0.64                        | 0.53               | 0.12        | 0.16        |

Figure 6: Comparative evaluation of PCTOM and SPCTOM trajectories.

Classical PCTOM solutions achieve minimum traversal time by operating directly on the feasibility boundary with bang-bang or bang-singular-bang control structures. While this minimizes nominal motion time, the resulting discontinuities in actuator torques produce impulsive jerk and excite structural vibrations in practical systems. Such trajectories are difficult to track accurately on physical manipulators due to actuator dynamics, structural flexibility, and modelling inaccuracies.

In contrast, SPCTOM explicitly constrains actuator torque rates, resulting in smooth pseudo-

velocity, pseudo-acceleration, and pseudo-jerk profiles. Although the inclusion of jerk constraints may increase the nominal traversal time, the smoother trajectories significantly improve tracking behaviour and reduce oscillations, actuator stress, and mechanical wear. Consequently, the effective execution time in practical systems may remain competitive due to reduced tracking mismatch and controller excitation.

## 5.1 Evaluation of the 1-DoF Car Problem

The implemented framework was further evaluated on the standard 1-DoF car problem.

The reimplemented SPCTOM framework successfully generated smooth trajectories satisfying the imposed jerk constraints. Compared to the classical PCTOM solution, the resulting pseudo-velocity and control profiles were continuous and free from impulsive switching behaviour.

However, the obtained SPCTOM trajectory exhibited slight oscillatory behaviour, primarily due to the spline parameterization and the sensitivity of the numerical optimization process. Furthermore, the traversal time increased from approximately 2 s for the PCTOM solution to approximately 7 s for the SPCTOM solution. This indicates that the optimization converged to a conservative feasible solution rather than a near-boundary optimal solution.

Despite the increased traversal time, the SPCTOM solution demonstrated the primary objective of the framework: generating dynamically smoother trajectories that are more physically realizable and less demanding on the actuator system.

## 5.2 Implementation Challenges

Several practical challenges were encountered during the implementation and evaluation of the SPCTOM framework.

A feasible initialization is essential for the optimization process. Although trivially feasible solutions can always be obtained using sufficiently slow trajectories, identifying initial solutions close to the feasibility boundary is significantly more difficult.

The Flexible Tolerance Method combined with Nelder-Mead optimization relies heavily on constraint violation measures. Consequently, the optimization performance is highly sensitive to tolerance parameters and penalty tuning. Poor parameter selection may produce overly conservative trajectories or imperfect constraint satisfaction.

The convergence speed and solution quality also depend strongly on initialization. In the implemented framework, the optimizer frequently converged slowly and occasionally became trapped in local minima, particularly near highly constrained regions.

Another major challenge arises from the dependence on switching points obtained from the PCTOM solution. The Shiller-Lu algorithm used for switching-point computation can become fragile near singular points and dynamic singularities, reducing overall robustness and occasionally causing instability in the optimization initialization.

These observations suggest that more robust switching-point detection techniques and improved constrained optimization methods are required for reliable operation near the feasibility boundary. In particular, optimization approaches capable of handling nonlinear feasibility constraints more effectively may significantly improve convergence quality and reduce conservative behaviour in the resulting SPCTOM trajectories.

## 6 Conclusion

This report presented a study of smooth and time-optimal trajectory planning for industrial manipulators under torque and torque-rate constraints. By incorporating actuator jerk limitations indirectly through torque-rate constraints, SPCTOM produces trajectories that are smoother and more practically realizable than classical PCTOM solutions.

The formulation reduces the manipulator dynamics into a scalar phase-plane optimization problem and employs spline parameterization together with numerical optimization methods to compute feasible trajectories.

Although the framework improves smoothness and tracking behavior, practical implementation remains sensitive to initialization, optimization tuning, and switching-point detection. Future work may focus on more robust optimization strategies and improved handling of singularities and feasibility boundaries.

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